

AMGPSU-I48L-P120

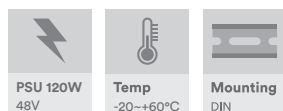
LIGHT INDUSTRIAL DIN-RAIL

120W POWER SUPPLY



Light Industrial Power Solutions

AMG's light industrial DIN-Rail 120W power supplies provide reliable power for AMG PoE based products and ensure stable equipment operation over a wide temperature range. They are suitable for all AMG PoE products (depending on voltage).



[AMGPSU-I48L-P120]

/ OVERVIEW

Designed in an ultra slim, robust DIN rail housing, the AMGPSU-I48L-P120 light industrial power supply is ideally suited for powering AMG PoE Ethernet equipment. Its wide operating temperature range ensures reliable operation in challenging environments.

Available in a 48-56V output version ensures the power supply is suitable for any PoE requirement.

The power supply offers a high level of stability and immunity to noise and a low ripple for best in class performance.

Compliant to the international IEC62368 standards for EMC and are safety approved to IEC/EN61000-4, CISPR32, EN55032, UL61010, IEC62368 and EN62368.

A wide voltage input range that features dual-use inputs for both DC and AC voltages that support 90-264V_{AC} or 127-370V_{DC} ensures the widest possible site support.

A range of other output power levels are available within the AMGPSU product range.

/ FEATURES

- Ultra slim size – ideal for confined spaces, including camera poles and roadside cabinets
- -20°C to +60°C temperature maintains performance in challenging environmental conditions
- DIN rail mountable – quick to install and remove for maintenance
- High efficiency - up to 89% typical
- Universal 90-264V_{AC} or 127-370V_{DC} input range
- Output short circuit, over-current and over-voltage protection included as standard
- High I/O isolation test voltage up to 4000V_{AC}
- Low ripple & noise
- EN62368 & UL safety approved
- AMG 3 Year Support Warranty

Specifications.

Input.

Characteristics	Operating Conditions	Min.	Typ.	Max.	Unit
Input Voltage Range	AC Input	90	-	264	VAC
	DC Input	127	-	370	VDC
Input Frequency		47	-	63	Hz
Input Current	115VAC	-	-	2.7	A
	230VAC	-	-	1.6	
Inrush Current	115VAC Cold Start	-	30	-	
	230VAC Cold Start	-	55	-	
Leakage Current	264VAC	<1.0mA			
Connector		3-Way Screw Terminal			

Output.

Characteristics	Operating Conditions	Min.	Typ.	Max.	Unit
Output Voltage Accuracy	Full Load Range	-	±1	-	%
Line Regulation	Rated Load	-	±0.5	-	
Load Regulation	0% - 100% Load	-	±1	-	
Output Ripple & Noise*	20MHz Bandwidth (peak-to-peak value)	-	-	150	mV
Temperature Coefficient		-	±0.03	-	%/°C
Short Circuit Protection	Recovery time <3s after the short circuit disappears	Constant Current, Continuous, Self-Recovery			
Over-Current Protection	230VAC Rated Load, Normal/High Temp	105%-150%Io, Constant Current, Automatic Recovery After Fault			
	230VAC Rated Load, Low Temp	≥105%Io, Constant Current, Automatic Recovery After Fault			
Over-Voltage Protection		≤60V (Output Voltage Turn Off, Re-Power Unit For Recovery)			
Over-Temperature Protect	Output Volatge Turn Off, Re-Power Unit For Recovery				
Minimum Load		0	-	-	%
Start-up Delay Time		-	-	3	s
Hold-up Time	115VAC	8	-	-	ms
	230VAC	16	-	-	
Connector		4-Way Screw Terminal			
Note: *The “tip and barrel method” is used for ripple and noise test, output parallel 47μF electrolytic capacitor and 0.1μF ceramic capacitor.					

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Specifications.

General.

Characteristics		Operating Conditions	Min.	Typ.	Max.	Unit
Isolation Test	Input-Earth	Electric Strength Test for 1 min., (leakage current <10mA)	2000	-	-	VAC
	Input-Output		4000	-	-	
	Output-Earth		500	-	-	
Insulation Resistance	Input-Earth	At 500 VDC	100	-	-	M Ω
	Input-Output		100	-	-	
	Output-Earth		100	-	-	
Operating Temperature			-20	-	+60	°C
Storage Temperature			-40	-	+85	
Operating Humidity		Non-Condensing	20	-	90	%RH
Storage Humidity			10	-	95	
Switching Fequency			-	65	-	kHz
Operating Temperature Power Derating		-20°C to -10°C (115VAC)	2.0	-	-	% / °C
		-20°C to -10°C (230VAC)	0	-	-	
		+40°C to +60°C (115VAC)	2.5	-	-	
		+50°C to +60°C (230VAC)	5	-	-	
Input Voltage Derating		90VAC to 115VAC	1	-	-	%/VAC
Safety Standard			UL61010/UL508/EN62368			
Safety Class			Class I			
MTBF		MIL-HDBK-217F @ 25°C	>300,000 hours			

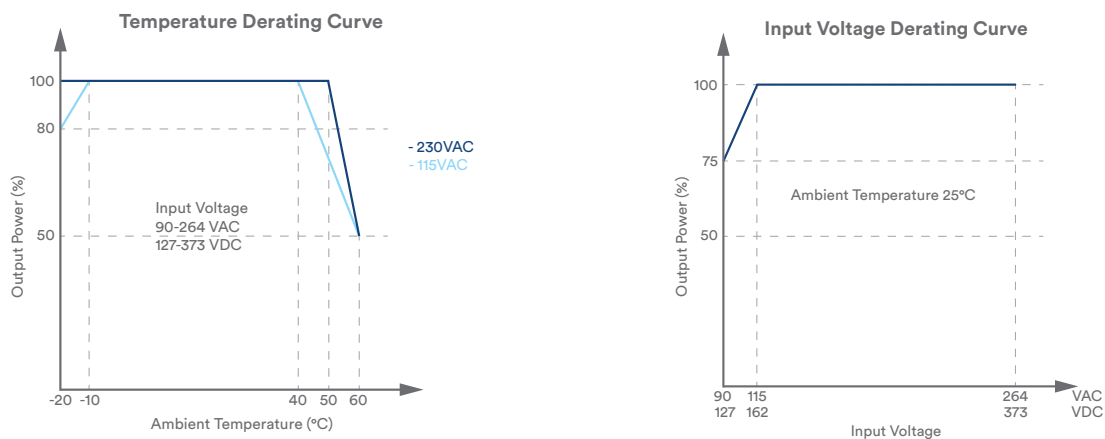
Mechanical.

Case Material	Aluminium
Dimensions	125 × 36 × 100 mm (4.92 × 1.42 × 3.94 in) (H x W x D)
Weight	0.41 Kg
Cooling	Free Air Convection

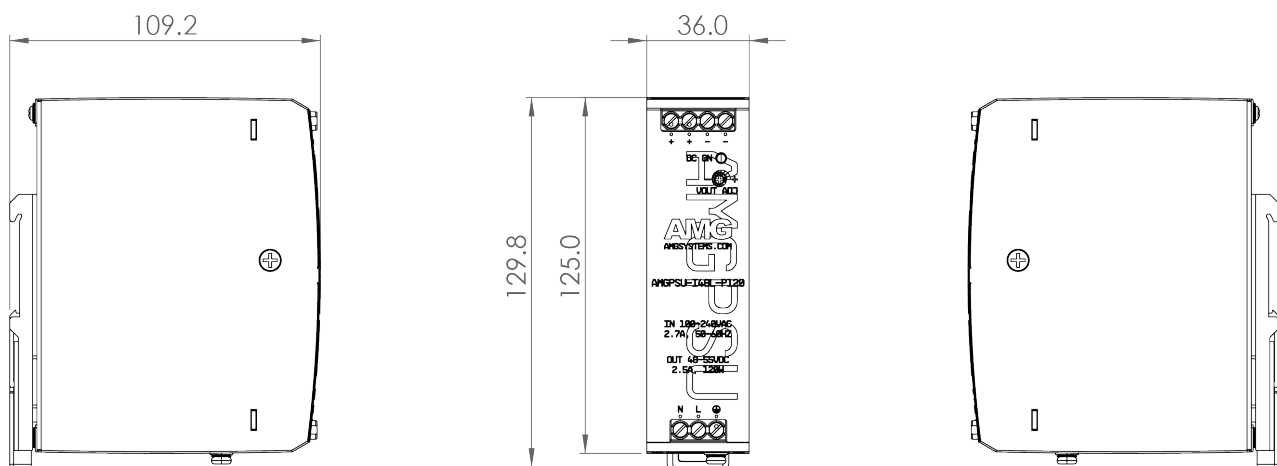
Regulatory.

Emissions	CE	CISPR32/EN55032 Class A
	RE	CISPR32/EN55032 Class A
	Harmonic Current	IEC/EN61000-3-2 Class A
Immunity	ESD	IEC/EN 61000-4-2 (Contact ±6KV / Air ±8KV)
	RS	IEC/EN 61000-4-3 (10V/m)
	EFT	IEC/EN 61000-4-4 (±4KV)
	Surge	IEC/EN 61000-4-5 (Line - Line ±2KV, Line - GND ±4KV)
	CS	IEC/EN 61000-4-6 (10V r.m.s)
	Voltage Dips, Short Interruptions and Voltage Variations Immunity	IEC/EN 61000-4-11 (0%, 70%)

Product Characteristic Curve.



Product Dimensions.



Part Numbers.

120W Light Industrial DIN-Rail Power Supplies

AMGPSU-I48L-P120	Light Industrial DIN Rail Power Supply, 48V Nominal Out (48-56V Adjustable), 120W (2.5A)
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Notes.

Unless otherwise specified, parameters in this datasheet were measured under the conditions of Ta=25°C. humidity <75% RH with nominal input voltage and rated output load.

In a continuing effort to improve and advance technology, product specifications are subject to change without notice. Please visit www.amgsystems.com for the latest product specifications.

